

Problem Set 2 Solutions Q1-Q3

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1 Population Regression vs CEF

In lecture 2, we considered three things: y_i , $E[y_i|x_i]$ and $x_i'\beta^*$ the "infeasible" or population OLS estimator. We showed that we can write:

$$y_i = x_i'\beta^* + u_i$$
$$\text{where } \beta^* = \operatorname{argmin}_b E[(y_i - x_i'b)^2] \quad (1)$$

1. Show that $E[x_i u_i] = 0$.

Answer: By FOC of the problem (1) above:

$$\frac{\partial}{\partial b} = -2E[x_i(y_i - x_i'b)] = 0 \text{ at } b = \beta^*$$
$$\Rightarrow E[x_i(y_i - x_i'\beta^*)] = E[x_i u_i] = 0. \quad (2)$$

2. Show $\beta^* = E[x_i x_i']^{-1} E[x_i y_i]$.

Answer: This could be shown from the FOC (2) above: $E[x_i(y_i - x_i'\beta^*)] = 0 \Rightarrow \beta^* = E[x_i x_i']^{-1} E[x_i y_i]$.

3. Show $E[u_i|x_i] = 0 \Rightarrow E[x_i u_i] = 0$. Hint: Apply the Law of Iterated Expectations.

Answer: If we have $E[u_i|x_i] = 0$, by the L.I.E.,

$$E[x_i u_i] = E_x[E[x_i u_i|x]]$$
$$= E_x[x_i E[u_i|x]]$$
$$= E_x[x_i * 0] = 0.$$

4. List two cases where $E[u_i|x_i] = 0$. Clearly state your assumption and explain why $E[u_i|x_i] = 0$ under such assumptions.

Answer: we show two cases in class where the conditional expectation function (CEF) is linear, in which case the residual from the population regression (1) satisfies $E[u_i|x_i] = 0$, as $E[y_i|x_i] = x_i'\beta^*$.

Case 1: x_i are dummies for a categorical variable. Suppose each unit is either in group A or B. Define $x_i = (D_i^{(a)}, D_i^{(b)})$, where $D_i^{(a)} = 1[i \in a]$ indicates unit i is in group a , and $D_i^{(b)} = 1[i \in b]$. Then we have population regression coefficients

$$\beta^* = \begin{bmatrix} E[y_i|D_i^{(a)} = 1] \\ E[y_i|D_i^{(b)} = 1] \end{bmatrix}$$
$$E[y_i|x_i] = D_i^{(a)} \times E[y_i|D_i^{(a)} = 1] + D_i^{(b)} \times E[y_i|D_i^{(b)} = 1]$$
$$= x_i'\beta^*.$$

Case 2: linear normal model - assume y conditional on x is normally distributed. Linear normal model assumes:

$$y_i = x_i'\beta + u_i, u_i \sim N(0, \sigma^2)$$

In this special case, u_i is independent of x_i , and satisfies $E[u_i|x_i] = 0$.

2 Regressions with Dummies

Suppose we have a sample of size N made up of two groups, denoted 0 and 1. Let D_i be a dummy variable with $D_i = 1$ indicating membership in group 1. Finally, let N_0 and N_1 represent the size of the two subgroups (so $N = N_0 + N_1$).

a) Consider an OLS regression model:

$$y_i = \alpha + \beta D_i + u_i$$

Using the first order conditions for the OLS estimates $(\hat{\alpha}, \hat{\beta})$, show that

$$\begin{aligned}\hat{\alpha} &= \frac{1}{N_0} \sum_{i \in 0} y_i, \\ \hat{\alpha} + \hat{\beta} &= \frac{1}{N_1} \sum_{i \in 1} y_i\end{aligned}$$

• **Answer:**

$$(\hat{\alpha}, \hat{\beta}) = \underset{a, b}{\operatorname{argmin}} \sum_{i=1}^N (y_i - a - b D_i)^2$$

First order conditions:

$$\frac{\partial}{\partial a} = 0 \Rightarrow \sum_{i=1}^N (y_i - \hat{\alpha} - \hat{\beta} D_i) = 0 \quad (3)$$

$$\frac{\partial}{\partial b} = 0 \Rightarrow \sum_{i=1}^N (y_i - \hat{\alpha} - \hat{\beta} D_i) D_i = 0$$

When $D_i = 0$, $(y_i - \hat{\alpha} - \hat{\beta} D_i) D_i = 0$,

$$\begin{aligned}\sum_{i=1}^N (y_i - \hat{\alpha} - \hat{\beta} D_i) D_i &= \sum_{i: D_i=1} (y_i - \hat{\alpha} - \hat{\beta}) = 0 \\ &\Rightarrow \frac{1}{N_1} \sum_{i: D_i=1} (y_i - \hat{\alpha} - \hat{\beta}) = 0 \\ &\Rightarrow \hat{\alpha} + \hat{\beta} = \frac{1}{N_1} \sum_{i: D_i=1} y_i.\end{aligned} \quad (4)$$

And the FOC with respect to a :

$$\begin{aligned}\sum_{i=1}^N (y_i - \hat{\alpha} - \hat{\beta} D_i) &= \sum_{i: D_i=0} (y_i - \hat{\alpha}) + \underbrace{\sum_{i: D_i=1} (y_i - \hat{\alpha} - \hat{\beta})}_{=0 \text{ by (4)}} \\ &= \sum_{i: D_i=0} (y_i - \hat{\alpha}) \\ &\Rightarrow \hat{\alpha} = \frac{1}{N_0} \sum_{i: D_i=0} y_i.\end{aligned}$$

b) Now consider a case where there are 3 groups: 0, 1, 2 and there are 2 dummies: $D_{1i} = 1$ if i is in group 1, and 0 otherwise, and $D_{2i} = 1$ if i is in group 2, and 0 otherwise, with the regression model:

$$y_i = \alpha + \beta_1 D_{1i} + \beta_2 D_{2i} + u$$

- i) find the first order conditions for minimizing the sum of squared residuals, which define the OLS estimator.

Answer:

$$(\hat{\alpha}, \hat{\beta}_1, \hat{\beta}_2) = \underset{a, b_1, b_2}{\operatorname{argmin}} \sum_{i=1}^N (y_i - a - b_1 D_{1i} - b_2 D_{2i})^2$$

First order conditions:

$$\begin{aligned} \frac{\partial}{\partial a} = 0 &\Rightarrow \sum_{i=1}^N (y_i - \hat{\alpha} - \hat{\beta}_1 D_{1i} - \hat{\beta}_2 D_{2i}) = 0 \\ \frac{\partial}{\partial b_1} = 0 &\Rightarrow \sum_{i=1}^N (y_i - \hat{\alpha} - \hat{\beta}_1 D_{1i} - \hat{\beta}_2 D_{2i}) D_{1i} = 0 \\ \frac{\partial}{\partial b_2} = 0 &\Rightarrow \sum_{i=1}^N (y_i - \hat{\alpha} - \hat{\beta}_1 D_{1i} - \hat{\beta}_2 D_{2i}) D_{2i} = 0 \end{aligned} \quad (5)$$

- ii) using the FOC show that the OLS estimators for the 3-group model will have

$$\begin{aligned} \hat{\alpha} &= \frac{1}{N_0} \sum_{i \in 0} y_i \\ \hat{\alpha} + \hat{\beta}_1 &= \frac{1}{N_1} \sum_{i \in 1} y_i \\ \hat{\alpha} + \hat{\beta}_2 &= \frac{1}{N_2} \sum_{i \in 2} y_i \end{aligned}$$

Answer: First, look at the FOC - (5), $(y_i - \hat{\alpha} - \hat{\beta}_1 D_{1i} - \hat{\beta}_2 D_{2i}) D_{2i} = 0$ unless $D_{2i} = 1$ or $i \in 2$. So the summation:

$$\begin{aligned} &\sum_{i=1}^N (y_i - \hat{\alpha} - \hat{\beta}_1 D_{1i} - \hat{\beta}_2 D_{2i}) D_{2i} \\ &= \sum_{i \in 2} (y_i - \hat{\alpha} - \hat{\beta}_2) = 0 \\ &\Rightarrow \hat{\alpha} + \hat{\beta}_2 = \frac{1}{N_2} \sum_{i \in 2} y_i \end{aligned}$$

Similarly, $\hat{\alpha} + \hat{\beta}_1 = \frac{1}{N_1} \sum_{i \in 1} y_i$. Finally, plugging both into the first FOC:

$$\begin{aligned} &\sum_{i=1}^N (y_i - \hat{\alpha} - \hat{\beta}_1 D_{1i} - \hat{\beta}_2 D_{2i}) \\ &= \sum_{i \in 0} (y_i - \hat{\alpha}) + \underbrace{\sum_{i \in 1} (y_i - \hat{\alpha} - \hat{\beta}_1)}_{=0} + \underbrace{\sum_{i \in 2} (y_i - \hat{\alpha} - \hat{\beta}_2)}_{=0} \\ &\Rightarrow \hat{\alpha} = \frac{1}{N_0} \sum_{i \in 0} y_i. \end{aligned}$$

3 Residualized Regression (Frisch-Waugh)

In lecture4, we showed that the j^{th} row of the population regression coefficient β^* from the model

$$y_i = x_i \beta^* + u_i$$

can be obtained by first getting the residual from an auxiliary regression of x_{ji} on all the other x' s:

$$x_{ji} = x'_{(\sim j)i} \pi + \xi_i \quad (6)$$

then forming:

$$\beta_j^* = E[\xi_i^2]^{-1} E[\xi_i y_i]$$

Suppose that we also “residualized” the dependent variable using an auxiliary regression of y_i on all the other x' s:

$$y_i = x'_{(\sim j)i} \lambda + \phi_i \quad (7)$$

Show that its also true that:

$$\beta_j^* = E[\xi_i^2]^{-1} E[\xi_i \phi_i]$$

(In other words, we could residualize BOTH x_{ji} AND y_i and still get the same right answer).

Answer: Start with the Frisch-Waugh Theorem, and plug in (7) $y_i = x'_{(\sim j)i} \lambda + \phi_i$:

$$\begin{aligned} \beta_j^* &= E[\xi_i^2]^{-1} E[\xi_i y_i] \\ &= E[\xi_i^2]^{-1} E[\xi_i (x'_{(\sim j)i} \lambda + \phi_i)] \\ &= E[\xi_i^2]^{-1} \left(E[\xi_i x'_{(\sim j)i} \lambda] + E[\xi_i \phi_i] \right) \end{aligned}$$

by the auxiliary regression (6): $E[\xi_i x'_{(\sim j)i} \lambda] = 0$ as the other regressors are orthogonal to the residual ξ_i , and λ is just a vector of constants:

$$E[\xi_i x'_{(\sim j)i} \lambda] = E[\xi_i x_{(\sim j)i}]' \lambda = 0.$$

Therefore

$$\beta_j^* = E[\xi_i^2]^{-1} E[\xi_i \phi_i]$$

which means you can also residualize the outcome over regressors x_{-j} , and then estimate β_j by regressing ϕ_i on ξ_i .